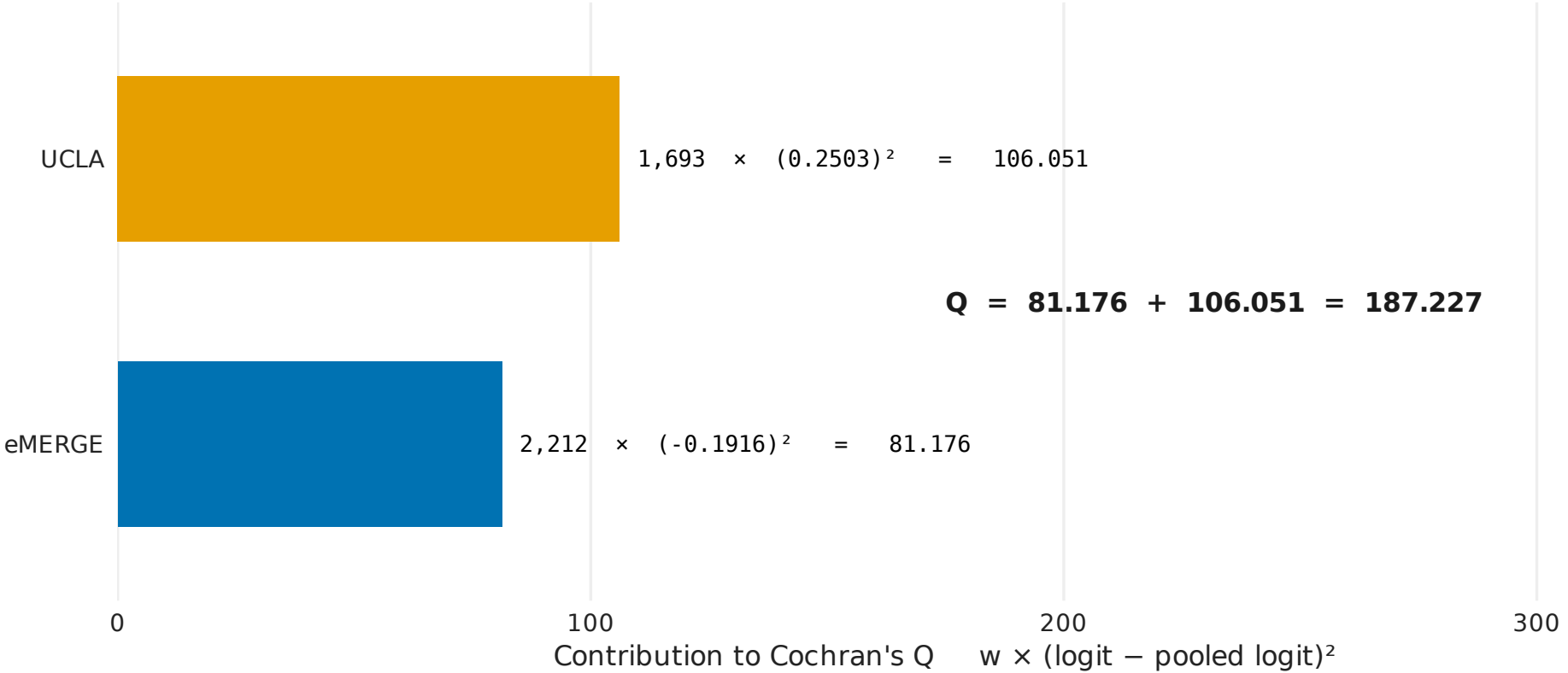
4. Cochran's Q is the sum of the weighted squared distances

Square each distance from the balance point in plot 3, so the sign drops out.
Multiply by that study's weight, then add the two products. That sum is Q.



Q = 187.227. It describes scatter around the pooled logit. The pooled AUC stays 0.645.
eMERGE contributes 81.176 and UCLA contributes 106.051. Squaring gives the study that sits further from the mean the larger piece.